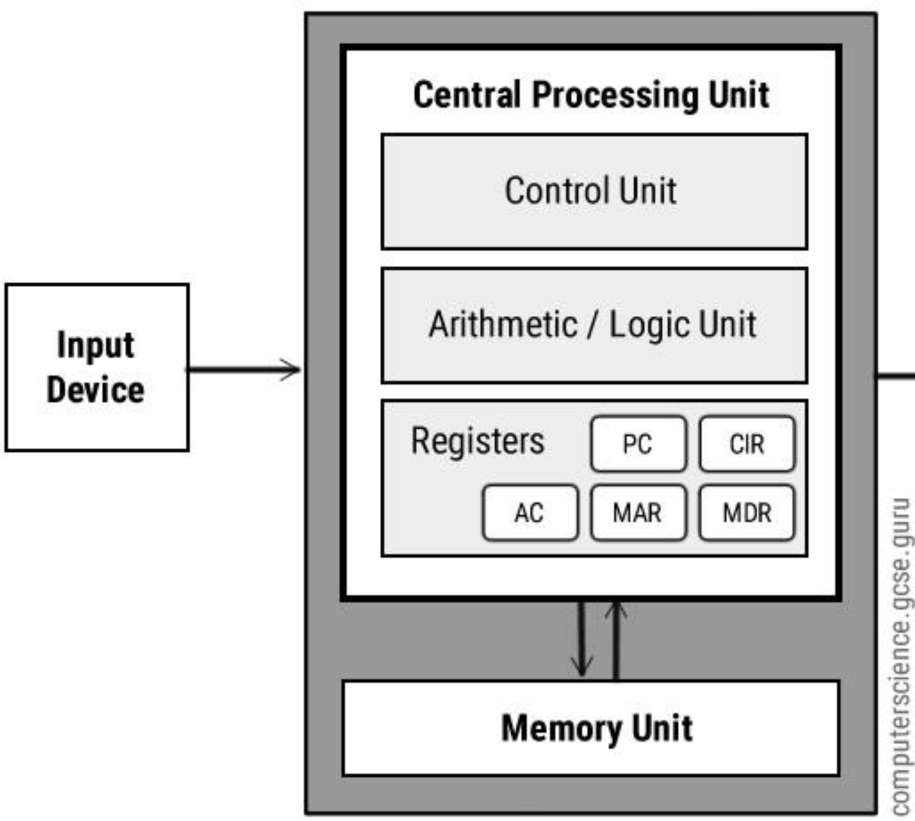
# Types of Computer Architecture:

# Von Neumann Architecture

Von Neumann architecture was first published by John von Neumann in 1945.

His computer architecture design consists of a Control Unit, Arithmetic and Logic Unit (ALU), Memory Unit, Registers and Inputs/Outputs.

Von Neumann architecture is based on the stored-program computer concept, where instruction data and program data are stored in the same memory.  This design is still used in most computers produced today.




# Central Processing Unit (CPU)

The Central Processing Unit (CPU) is the electronic circuit responsible for executing the instructions of a computer program.

It is sometimes referred to as the microprocessor or processor.

The CPU contains the ALU, CU and a variety of registers.

## Registers

Registers are high speed storage areas in the CPU. All data must be stored in a register before it can be processed.

| | | |
|---|---|---|
| **MAR** | Memory Address Register | Holds the memory location of data that needs to be accessed |
| **MDR** | Memory Data Register | Holds data that is being transferred to or from memory |
| **AC** | Accumulator | Where intermediate arithmetic and logic results are stored |
| **PC** | Program Counter | Contains the address of the next instruction to be executed |
| **CIR** | Current Instruction Register | Contains the current instruction during processing |

## Arithmetic and Logic Unit (ALU)

The ALU allows arithmetic (add, subtract etc) and logic (AND, OR, NOT etc) operations to be carried out.

## Control Unit (CU)

The control unit controls the operation of the computer's ALU, memory and input/output devices, telling them how to respond to the program instructions it has just read and interpreted from the memory unit.

The control unit also provides the timing and control signals required by other computer components.

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# Buses

Buses are the means by which data is transmitted from one part of a computer to another, connecting all major internal components to the CPU and memory.

A standard CPU system bus consists of a control bus, data bus and address bus.

| | |
|---|---|
| **Address Bus** | Carries the addresses of data (but not the data) between the processor and memory |
| **Data Bus** | Carries data between the processor, the memory unit and the input/output devices |
| **Control Bus** | Carries control signals/commands from the CPU (and status signals from other devices) in order to control and coordinate all the activities within the computer |

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# Memory Unit

The memory unit consists of RAM, sometimes referred to as primary or main memory.  Unlike a hard drive (secondary memory), this memory is fast and also directly accessible by the CPU.

RAM is split into partitions.  Each partition consists of an address and its contents (both in binary form).

The address will uniquely identify every location in the memory.

Loading data from permanent memory (hard drive), into the faster and directly accessible temporary memory (RAM), allows the CPU to operate much quicker.

# Harvard Architecture

Harvard architecture is a computer design model where program instructions and data are stored in separate memory units that are accessed through independent buses. This separation allows the processor to fetch instructions and access data simultaneously, which helps avoid the bottleneck present in traditional Von Neumann systems.

- Eliminates the Von Neumann bottleneck.
- Faster and predictable performance (suitable for real-time systems).
- Parallel access to both instructions and data.

## Working Principle

In Harvard Architecture, fetching an instruction from instruction memory and reading/writing data from/to data memory happen at the same time without waiting for one to finish. Separate buses prevent the bottleneck that occurs when data and instructions share a path. For example, while an instruction is being executed, the next instruction can be fetched simultaneously, speeding up processing.

## Buses

Buses are used as signal pathways. In Harvard architecture, there are separate buses for both instruction and data. Types of Buses:

- Data Bus: It carries data among the main memory system, processor, and I/O devices.

- Data Address Bus: It carries the address of data from the processor to the main memory system.
- Instruction Bus: It carries instructions among the main memory system, processor, and I/O devices.
- Instruction Address Bus: It carries the address of instructions from the processor to the main memory system.

## Components of Harvard Architecture

Harvard architecture is designed with specific components that handle instruction execution, control, and data communication.

- Arithmetic and Logic Unit: The arithmetic logic unit is part of the CPU that operates all the calculations needed. It performs addition, subtraction, comparison, logical Operations, bit Shifting Operations, and various arithmetic operations.
- Control Unit: The Control Unit is the part of the CPU that operates all processor control signals. It controls the input and output devices and also controls the movement of instructions and data within the system.
- Input/Output System: Input devices are used to read data into main memory with the help of CPU input instruction. The information from a computer as output is given through Output devices. The computer gives the results of computation with the help of output devices.

## Application of Harvard Architecture:

Harvard architecture is a type of computer design where the memory for instructions and data are kept separate. Here are the some applications:

Digital Signal Processors (DSPs):

- Audio and video processing, telecommunications, radar systems, and image processing.
- Texas Instruments TMS320 for hearing aids.

Microcontrollers (MCUs):

- Embedded systems in consumer electronics, automotive systems, IoT devices, and industrial automation.
- PIC in automotive ABS

Network Processors:

- Routers, switches, and network security appliances.
- Broadcom StrataXGS

Automotive Systems:

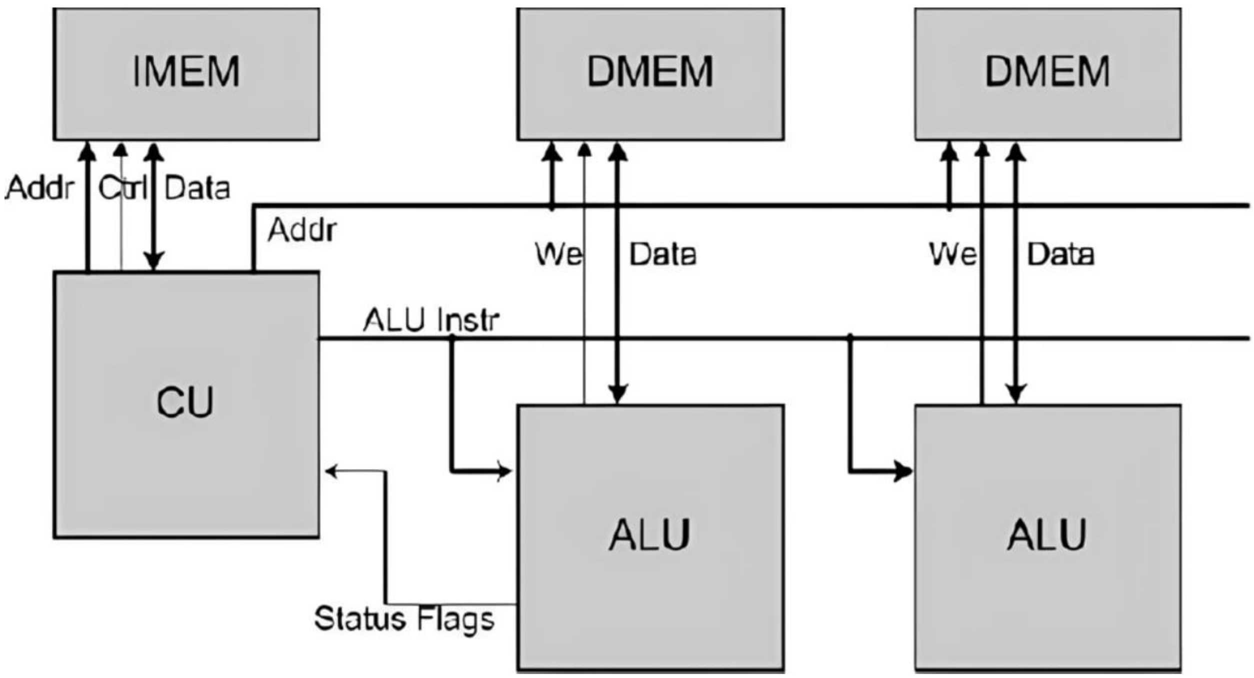
- Engine control units (ECUs), advanced driver-assistance systems (ADAS), and infotainment systems.
- NXP S32K for engine control

# 3. Modified Harvard Architecture

**A Modified Harvard Architecture is a hybrid type of computer architecture that combines features of both the classic Harvard architecture and the more widely used von Neumann architecture.**

**Like a true Harvard architecture, a modified Harvard architecture utilizes separate caches for instructions and data. These caches are much faster than main memory, so frequently accessed instructions and data can be retrieved quickly.**

However, unlike the pure Harvard architecture where instructions and data have completely separate physical memory units, a modified Harvard architecture keeps instructions and data in the same main memory



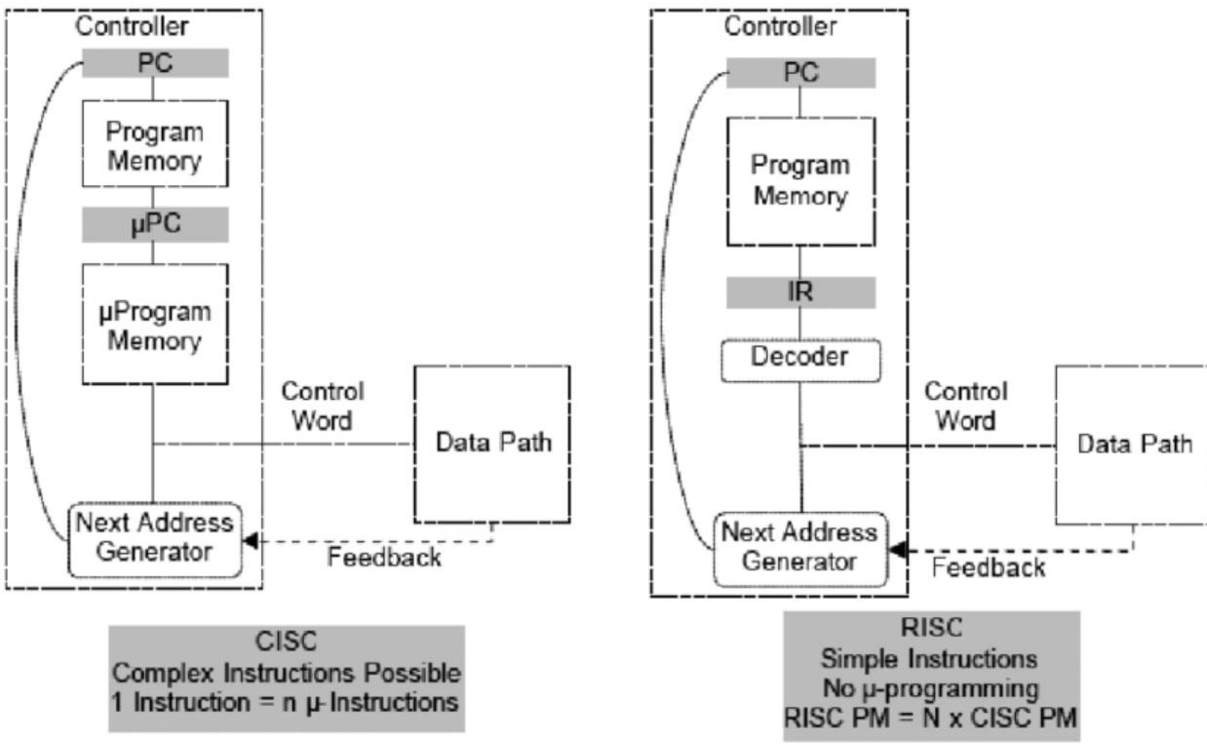

This combination allows for simultaneous access to instructions and data, boosting performance over a standard von Neumann architecture with a single cache. Compared to a true Harvard architecture with separate memory units, the unified memory simplifies the design and reduces costs.

Many processors you'll find in computers today use a modified Harvard architecture with separate instruction and data caches

# 4. RISC & CISC Architectures

RISC (Reduced Instruction Set Computing) and CISC (Complex Instruction Set Computing) are two different architectures for computer processors that determine how they handle instructions.

RISC processors are designed with a set of basic, well-defined instructions that are typically fixed-length and easy for the processor to decode and execute quickly. The emphasis in RISC is on designing the hardware to execute simple instructions efficiently, leading to faster clock speeds and potentially lower power consumption. Examples of RISC processors include ARM processors commonly found in smartphones and tablets, and MIPS processors used in some embedded systems.




CISC processors, however, have a wider range of instructions, including some very complex ones that can perform multiple operations in a single instruction. This can be more concise for programmers but can take the processor more time to decode and execute.

The goal of CISC is to provide a comprehensive set of instructions to handle a wide range of tasks, potentially reducing the number of instructions a programmer needs to write. Examples of CISC processors include Intel’s x86 processors, which are used in most

**personal computers, and Motorola 68000 family of processors which are used in older Apple computers.**

# Components of Computer Architecture

**While computer architectures can differ greatly depending on the purpose of the computer, several key components generally contribute to its structure. These include:**

1. **Central Processing Unit (CPU) - Often referred to as the "brain" of the computer, the CPU executes instructions, performs calculations, and manages data. Its architecture dictates factors such as instruction set, clock speed, and cache hierarchy, all of which significantly impact overall system performance.**

2. **Memory Hierarchy - This includes various types of memory, such as cache memory, random access memory (RAM), and storage devices. The memory hierarchy plays a crucial role in optimizing data access times, as data moves between different levels of memory based on their proximity to the CPU and the frequency of access.**

3. **Input/Output (I/O) System - The I/O system enables communication between the computer and external devices, such as keyboards, monitors, and storage devices. It involves designing efficient data transfer mechanisms to ensure smooth interaction and data exchange.**

4. **Storage Architecture - This deals with how data is stored and retrieved from storage devices like hard drives, solid-state drives (SSDs), and optical drives. Efficient storage architectures ensure data integrity, availability, and fast access times.**

5. **Instruction Pipelining - Modern CPUs employ pipelining, a technique that breaks down instruction execution into multiple stages. This allows the CPU to process multiple instructions simultaneously, resulting in improved throughput.**

6. **Parallel Processing - This involves dividing a task into smaller subtasks and executing them concurrently, often on multiple cores or processors. Parallel processing significantly accelerates computations, making it key to tasks like simulations, video rendering, and machine learning.**

**All of the above parts are connected through a system bus consisting of the address bus, data bus and control bus.**